EE2016: Microprocessor theory and lab

Tutorial-1 (TGV)

Q1. Which addressing mode uses operands that are part of the instruction itself?

A) Direct

B) Register

C) Immediate

D) Indirect

Q2. In `MOV AX, [BX]`, the addressing mode for the source operand is:

A) Register Indirect

B) Direct

C) Immediate

D) Indexed

[BX] indicates the address pointed by the BX register.

Q3. The addressing mode where the operand address is explicitly given in the instruction is:

A) Immediate

B) Direct

C) Register Indirect

D) Implied

Q4. For the instruction `PUSH AX`, the addressing mode used for the operand `AX` is:

A) Implied

B) Register

C) Direct

D) Immediate

Q5. In indexed addressing mode, the effective address is calculated as:

A) Base Register + Index Register

B) Index Register + Displacement

C) Program Counter + Displacement

D) Stack Pointer + Offset

Q6. Relative addressing mode is primarily used for:

A) Data transfer

B) Branch instructions

C) Arithmetic operations

D) I/O operations

Q7. The addressing mode where the operand is implied by the instruction (e.g., **STC**) is:

A) Immediate
B) Direct
C) Implied
D) Register

The STC instruction sets the carry flag.

Q8. For MOV AX, [BP+10], the source operand uses:

A) Base-Indexed
B) Based Relative
C) Indexed
D) Register Indirect
BP is the base pointer register.

Q9. Consider an 8086 system where:

DS = 4000H

BX = 0200H

DI = 0040H

Memory location 40240H contains 1A2BH.

The instruction MOV AX, [BX+DI+10H] is executed. What value is loaded into AX?

A) 0200H
B) 0040H
C) 1A2BH
D) 40240H

Q10. In a processor with Base-Register Addressing, if the base register contains 3000H and the instruction

specifies a displacement of +200H, what is the effective address? What happens if the base register is

incremented to 3002H before instruction execution?

A) 3200H, 3202H

B) 3200H, 3200H

C) 3000H, 3002H

D) 3200H, 3202H with auto-indexing

Q11. Consider a 32-bit processor where:

EAX = 00001000H

EBX = 00002000H

Memory location 00003000H contains 00004000H

After MOV ECX, [EAX+EBX*2] , what is ECX?
A) 00003000H
B) 00005000H
C) Content of 00005000H
D) Content of 00003000H

Q12. In a RISC-V processor, the instruction 1d t0, 8(s1) uses which addressing mode? What is the role of s1 and 8?

A) Register Indirect; Base = s1, Offset = 8

B) PC-Relative; PC = s1, Offset = 8

C) Indexed; Index = s1, Base = 8

D) Immediate; Value = 8

Refer to the RISC-V instruction set using the internet to answer this question.

Q13. In an 8085 system, LDAX B uses:

A) Register Indirect
B) Direct
C) Immediate
D) Implied

Refer to Intel 8085 instruction set using the internet to answer this question.

Q14. Which addressing mode cannot be used for destination operands in typical ISAs?

A) Immediate
B) Register
C) Direct
D) PC-Relative

Q15. In an 8086 system, the following instructions are executed:

Assembly:

MOV AX, 1234H

MOV [2000H], AX

LES DI, [2000H]

If DS = 0100H, what are the values of ES and DI after execution?

A) ES = 1234H, DI = 0100H

B) ES = 0100H, DI = 1234H

C) ES = 1234H, DI = 5678H

D) ES = 5678H, DI = 1234H

Q16. Consider a microprocessor with a 20-bit address port. Suppose the opcode field takes 8 bits, then

What will be the length of an instruction if it follows (a) 4-address instruction format, (b) 2-address instruction format.